

SAM VERSCHOOREN, PhD

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Belgian (Belgium), Oct. 9 1991, M

<https://samverschooren.github.io/>**Education**

- 2016-2020:** Ph.D. in Psychology, Ghent Univ.
2014-2016: MA in Theoretical and Experimental Psychology (*magna cum laude*), Ghent Univ.
2012-2014: Prep. Course for MA Psychology (Th. & Exp.) (*magna cum laude*), Ghent Univ.
2012-2014: MA in Philosophy (*cum laude*), Ghent Univ.
2009-2012: BA in Philosophy (*cum laude*), Univ. of Antwerp

Academic experience

- 2025-** Postdoctoral fellow (MSCA) UiT – The Arctic University of Tromsø, NO
2024-2025 Postdoctoral fellow (AvH) Humboldt University & Max Planck Institute for Human Cognitive and Brain Sciences, DE
2023-2024: Visiting researcher (FWO/CWO), Max Planck Institute for Human Cognitive and Brain Sciences, DE
2021-2022: Postdoctoral fellow (BAEF), Duke University, USA
2020-2024: Postdoctoral fellow (FWO), Ghent University, BE
2019: Visiting researcher (FWO), Duke University, USA
2016-2020: PhD researcher, Ghent University, BE
2015-2016: Research internship, Netherlands Institute for Neuroscience, NL
2014: Research assistant, Ghent University, BE

Publications

- Verschooren, S.**, Dahl, M., Aly, M., Mittner, M. (in press) Transition dynamics of external and internal attention across on- and off-task states. *Nature Reviews Psychology*
- Wang, N., **Verschooren, S.**, Pourtois, G. (in press) Too alert to think straight: combined neurophysiological and modeling evidence of impaired internal shielding during hypervigilance. *Psychophysiology*
- Verschooren, S.**, Boeve, S., Vermeylen, L., Pourtois, G. (2025) Perceptual decoupling underlies internal shielding benefit during switches between external and internal attention: Evidence from early sensory ERP components. *Journal of Cognitive Neuroscience*
- Kessler, Y., **Verschooren, S.** (2025) Working memory updating for free items and for item-to-context bindings: When attention is enough and when gating is needed. *Memory and Cognition*
- Van der Biest, M., **Verschooren, S.**, Verbruggen, F., Brass, M. (2025) Perceptual judgements are resistant to the advisor's perceived level of trustworthiness: a deep fake approach. *Plos One*
- Van der Biest, M., **Verschooren, S.**, Verbruggen, F., Brass, M. (2025) Don't Judge a Book by its Cover: The Effect of Perceived Facial Trustworthiness on Advice Following in the Context of Value-Based Decision-Making. *Journal of Experimental Social Psychology*, 118, 104719
- Wang, N., **Verschooren, S.**, Vermeylen, L., Grahek, I., Pourtois, G. (2025) Hypervigilance strikes a balance between external and internal attention: behavioral and modeling evidence from the switching attention task. *Psychological Research*, 89 (1), 3
- Verschooren, S.**, Gaebler, M., Brass, M. (2024) From interoception to control over the internal body: The ideomotor hypothesis of voluntary interoaction. *Psychological Review*
- Verschooren, S.**, Egner, T. (2023) When the mind's eye prevails: the internal dominance over external attention (IDEA) hypothesis. *Psychonomic Bulletin & Review*, 1-21.
- Hautekiet, C., **Verschooren, S.**, Langerock, N., Vergauwe, E. (2023) Attentional Switching Between Perception and Memory: Examining Asymmetrical Switch Costs. *Attention, Perception, & Psychophysics*, 1-11.
- Verschooren, S.**, Schindler, S., De Raedt, R., Pourtois, G (2021) Early reduction of sensory processing within the extrastriate visual cortex when switching from internal to external attention. *Biological Psychology*, 163, 108119.
- Verschooren, S.**, Kessler, Y., Egner, T (2021) Evidence for a single mechanism gating perceptual and long-term memory information into working memory. *Cognition*, 212, 104668.

- Trutti, A. C., **Verschooren, S.**, Forstmann, B. U., Boag, R. (2021) Understanding subprocesses of working memory through the lens of model-based cognitive neuroscience. *Current Opinion in Behavioral Sciences*. 38, 57-65.
- Verschooren, S.**, Pourtois, G., Egner, T. (2020) More efficient shielding for internal than external attention? Evidence from asymmetrical switch costs. *Journal of Experimental Psychology: Human Perception and Performance*, 46(9), 912-925.
- Verschooren, S.**, Liefoghe, B., Brass, M., Pourtois, G. (2019) Attentional flexibility is imbalanced: Asymmetric cost for switches between external and internal attention. *Journal of Experimental Psychology: Human Perception and Performance*, 45(10), 1399-1414.
- Verschooren, S.**, Schindler, S., De Raedt, R., Pourtois, G. (2019) Switching attention from internal to external information processing: a review of the literature and empirical support of the resource sharing account. *Psychonomic Bulletin and Review*, 26(2), 468-490.
- Han, Y., Bruls, R., Soyman, E., Thomas, R.J., Pentaraki, V., Jelinek, N., Heinemans, M., Bassez, I., **Verschooren, S.**, Pruis, I., Van Lierde, T., Carrillo, N., Gazzola, V., Carrillo, M., Keyers, C. (2019) Bi-directional cingulate-dependent danger information transfer across rats. *PLOS Biology*, 17(12).

Preprint/under review

- Verschooren, S.**, Egner, T., Aly, M. (2026) Competition, concurrence, and cooperation between external and internal attention. https://osf.io/preprints/psyarxiv/ry94x_v1
- Wang, N., **Verschooren, S.**, Gasparello, A., Pourtois, G. (2026) Don't worry: anxious apprehension preserves the enhanced shielding of memory during attentional flexibility. https://osf.io/preprints/psyarxiv/nb9q6_v1
- Wang, N., **Verschooren, S.**, Pourtois, G. (2025) Dynamical contributions of reconfiguration and internal shielding to attentional flexibility: evidence from a drift diffusion model. <https://osf.io/preprints/psyarxiv/mdbf5>
- Liao, J., **Verschooren, S.**, Grahek, I., Mo, L., Pourtois, G. (2024) Value-driven Attentional Capture is Stronger for Internal than External Stimuli. <https://psyarxiv.com/e52n4/>

In preparation

- Verschooren, S.**, Agrawal, N., Fourcade, A., Gerosa, M., Mastek, H., Gaebler, M., Brass, M. (in prep) Ideomotor control over the internal body through autogenic training: a six-week pilot study
- Verschooren, S.**, Aasen, S. R., Mittner, M. (in prep) Mind wandering during external and internal task-focused states
- Aursand, M., Gaebler, M., Mittner, M., **Verschooren, S.** (in prep) The interoceptive flexibility task: Quantifying costs for switching to interoception from an external vs. internal attentional state
- Weinhardt, S., Dahl, M. J., Gaebler, M., Mittner, M., **Verschooren, S.** (in prep) The impact of interoceptive rhythms on attentional states and transitions
- Borovaya, M.*, Zanardi, T.*, Bhanot, S., **Verschooren, S.** (in prep) A novel framework for the cognitive science of yoga
- Barbot, L., **Verschooren, S.**, Senoussi, M. (in prep) To err is human: an exploration of error types in a visual working memory task and their relationship with clinical traits

Fellowships

Marie Skłodowska-Curie Actions

Postdoctoral fellowship (2 years), *UiT – The Arctic University of Norway* (start: 1 Aug 2025)
Approx. €220,000

Alexander von Humboldt Stiftung

Postdoctoral fellowship (2 years), *MPI CBS/Humboldt University* (start: 1 Oct 2024)
Approx. €110,000

Fonds Wetenschappelijk Onderzoek

Postdoctoral fellowship (3 years), *Ghent University* (start: 1 Oct 2020)
Approx. €320,000

Belgian-American Educational Foundation

Postdoctoral fellowship (1 year), *Duke University* (start: 1 Oct 2020)
Approx. \$50,000

Grants

Auroa Outstanding: Operating funds for the organization of a scientific colloquium (20,000 NOK)
WASP-HS: Understanding and Mitigating AI Manipulation: A Collaborative Workshop Series (498,000 SEK)
FWO: Mobility grant for research stay (8 months), MPI CBS (start: 1 June 2023)
CWO: Mobility grant for research stay (3 months), MPI CBS (start: 1 Feb 2023)
FWO: Mobility grant for research stay (5 months), Duke University (start: 1 Feb 2019)

Oral presentations (selected):

Verschooren, S. (2025) Transition dynamics of external and internal attention across on- and off-task states. *19th NVP Winter Conference on Brain and Cognition*, Egmond-aan-Zee, NL
Verschooren, S., (2025) Charting the flux of attention: transition dynamics of external and internal attention across on- and off-task states. *Invited talk at the lifespan psychology colloquium, Max Planck Institute for Human Development*, DE
Verschooren, S., (2024) From interoception to control over the internal body: The ideomotor hypothesis of voluntary interoaction. *Invited talk at the research colloquium, University of Wuerzburg*, DE
Verschooren, S. (2022) When the mind's eye prevails: The Internal Dominance over External Attention (IDEA) hypothesis. *Invited talk at the research colloquium, VU, Amsterdam*, NE
Verschooren, S. (2022) When the mind's eye prevails: The Internal Dominance over External Attention (IDEA) hypothesis. *Invited talk at the research seminar, University of Tromsø*, Tromsø, NO
Verschooren, S., Kessler, Y. (2022) The reference-back task revisited: Evidence for automatic WM updating from an ITI manipulation. *22nd Conference of the European Society for Cognitive Psychology*, Lille, FR
Verschooren, S., Pourtois, G., Egner, T. (2022) More efficient shielding for internal than external attention? Evidence from asymmetrical switch costs. *22nd Conference of the European Society for Cognitive Psychology*, Lille, FR
Verschooren, S. (2021) External and internal switch costs: Insights from the drift diffusion model. *Invited talk in Birte Forstmann's lab*. Amsterdam, NL
Verschooren, S., Pourtois, G., Egner, T. (2019) More Efficient Shielding for Internal than External Attention: Evidence from Asymmetrical Switch Costs. *17th NVP Winter Conference on Brain and Cognition*, Egmond-aan-Zee, NL
Verschooren, S. (2019) Switching between external and internal attention: declarative versus procedural working memory. *Invited talk in Evie Vergauwe's lab*. Genève, CH
Verschooren, S. (2019) Modeling interactions between perception, memory, and working memory. *Invited talk in Klaus Oberauer's lab*. Zurich, CH
Verschooren, S., Liefoghe, B., Marcel, B., Pourtois, G. (2019) Attentional Flexibility is Imbalanced: Asymmetric Cost for Switches between External and Internal Attention. *21th Conference of the European Society for Cognitive Psychology*, Tenerife, ES
Verschooren, S. (2019) BrainHack results: Drift diffusion modeling of working memory gating. *Model-based Neuroscience BrainHack*, Amsterdam, NL
Verschooren, S., Liefoghe, B., Marcel, B., Pourtois, G. (2019) Attentional Flexibility is Imbalanced: Asymmetric Cost for Switches between External and Internal Attention. *Data Blitz session CCN Retreat*, Durham, NC, USA

Poster presentations (selected)

Verschooren, S., Kessler, Y. Egner, T. (2019) Working Memory Gating for Perception and Long-Term Memory: Opening as a Serial and Closing as a Parallel Process. *7th Symposium on Motivation and Cognitive Control*, Berlin, DE
Verschooren, S., Kessler, Y. Egner, T. (2019) Modeling Working Memory Gating for External and Internal Information. *Model-based Neuroscience Summer School*, Amsterdam, NL
Verschooren, S., Liefoghe, B., Marcel, B., Pourtois, G. (2019) Attentional flexibility is imbalanced: Asymmetric Cost for Switches between External and Internal Attention. *1st Smokies Cognition & Affective Neuroscience Symposium*, Asheville, NC, USA
Verschooren, S., Schindler, S., De Raedt, R., Pourtois, G. (2018) Attention from inside out: P1 effects of shifts between internally and externally oriented attention. *Annual Meeting of Cognitive Neuroscience Society*, March 24-27, Boston, MA, USA

Skills

Experimental design	Javascript, Python, E-Prime, Matlab
Data analysis	Bayesian and frequentist multilevel modeling (<i>R</i>), Drift diffusion modeling (<i>R</i>), RM ANOVA (<i>JASP/R</i>), ERP (<i>Python</i>)
Communication of scientific findings (<i>written and presented</i>)	

Teaching

2025-2026:	Mind, Knowledge, and Reality (3 rd BA Philosophy, UiT)
2023-2024:	Attention in cognitive science and contemplative traditions (MA Mind & Brain, HU)
2023-2024:	Neuropsychologie (3 rd BA Physiotherapy, UGhent)
2020-2021:	Cognitive Psychology 1&2 (1 st and 2 nd MA Psychology, UGhent)
2019-2020:	Academic Skills – module Bias (2 nd BA Psychology, UGhent)
2017-2018:	Academic Skills – module Bias (2 nd BA Psychology, UGhent)

Supervision

PhD: Mathias Van Der Biest (UGent, 2024), Nan Wang (UGent, ongoing)
MA: Stefan Weinhardt (ongoing), Mathilde Aursand (ongoing), Helena Mastek (2024), Juliette Tacquet (2018), Gert Klügel (2019), Maël Reubens (2019), Hannelore Petitjean (2021), Gabrielle Van Tomme (2021)

Ad-hoc journal reviewing

Psychological Review; Journal of Experimental Psychology: Human Perception and Performance; Memory & Cognition; Attention, Perception, & Psychophysics; Nature Communications; Psychological Research; Consciousness & Cognition; Quarterly Journal of Experimental Psychology; Cognitive Processing; Physiology & Behavior; Journal of Cognitive Enhancement; BMC Neuroscience; ...

Ad-hoc grant reviewing

Austrian Science Fund (FWF; ESPRIT Program)

Organization of scientific meetings

Experimental Health Research Conference 2026, Tromsø, NO
 Cognitive Science of Yoga meeting, 2025, Berlin, DE

Conferences and Workshops

Conferences:

17th NVP Winter Conference on Brain and Cognition, Egmond-aan-Zee, NL
 21st Conference of the European Society for Cognitive Psychology, September 25-28, Tenerife, ES
 7th Symposium on Motivation and Cognitive Control, September 16-18, 2019, Berlin, DE
 1st Smokies Cognition & Affective Neuroscience Symposium, April 11-13, 2019, Asheville, NC, USA
 25th Annual Meeting of the Cognitive Neuroscience Society, March 24-27, 2018, Boston, MA, USA

Workshops/courses:

Neuroscience School of Advanced Studies: Interoception and Brain Disorders, May 10-17 2025, Venice, IT
 6th European Summer School for Mathematical and Computational Modeling of Cognition
 Advanced Academic English – Writing Skills, October 1 – December 16, Ghent, BE
 Model-based Neuroscience Summer School and Brainhack, August 5-14, Amsterdam, NL
 3rd CuttingEEG Symposium, June 19-22, 2017, Glasgow, UK
 EPOS Attention Workshop, November 24-25, 2016, Amsterdam, NL

Active collaborations

Tobias Egner, Duke University, USA
 Gilles Pourtois, Ghent Univ., BE

Mariam Aly, UC Berkeley, USA

Marcel Brass, Humboldt Univ., NE

Michael Gaebler, Max Planck Institute for Human Cognitive and Brain Sciences, DE

Matthias Mittner, University of Tromsø, NO

Martin Dahl, Max Planck Institute for Human Development, DE

Claire O'Callaghan, University of Sydney, AU

Professional Memberships

2022-2023: European Society for Cognitive Psychology

2021-2022: European Society for Cognitive Psychology

2021: Psychonomics

2019-2020: Belgian Association for Psychological Science

2017-2018: Cognitive Neuroscience Society

Public Outreach

Dag van de Wetenschap: Change blindness experiment (22 November 2020)

Science Norway: [Techniques to stay focused in the war for attention](#) (23 April 2026)